UNDERSTANDING EVOLUTION

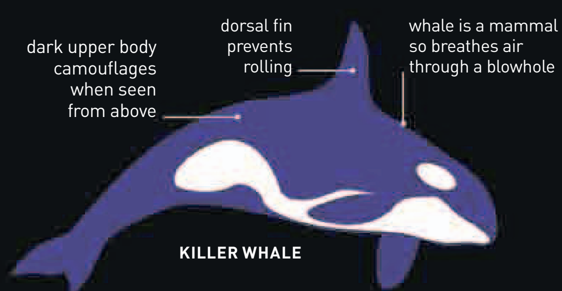
THE SOURCE OF BIOLOGICAL DIVERSITY, EVOLUTION ALSO PROVIDES A LINK TO OUR PREHISTORIC ANCESTORS

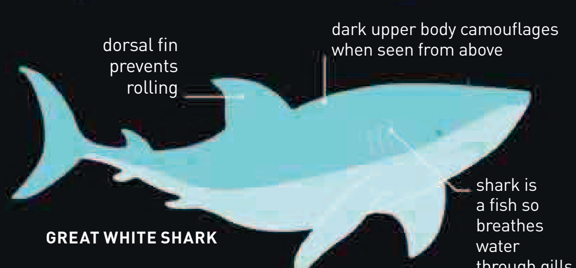
Fossils reveal how prehistoric life forms were different from those of today, and studies of different species suggest that all life originated from a single, simple ancestor that lived billions of years ago. Today, scientists understand the biological and genetic processes behind the evolutionary changes that gave rise to the diversity of life.

In the early 1800s, the French naturalist Jean-Baptiste Lamarck suggested, erroneously, that features acquired during an organism's lifetime could be passed on to its offspring. Later, Charles Darwin (see 1859) proposed that individuals are born with variations that make some of them "fitter"—more likely to survive and pass on their characteristics. This is called evolution by natural selection. It is now known that characteristics are determined by genes and that random gene mutations cause variation (see pp.284–85). But only natural selection can explain how some variations are better adaptations to the environment and come to predominate in organisms.

CONVERGENT EVOLUTION

Physical resemblance can indicate common ancestry but sometimes completely unrelated species evolve independently to become alike. Known as convergent evolution, this often happens when species have comparable roles in the same environment, so natural selection works on them in similar ways.



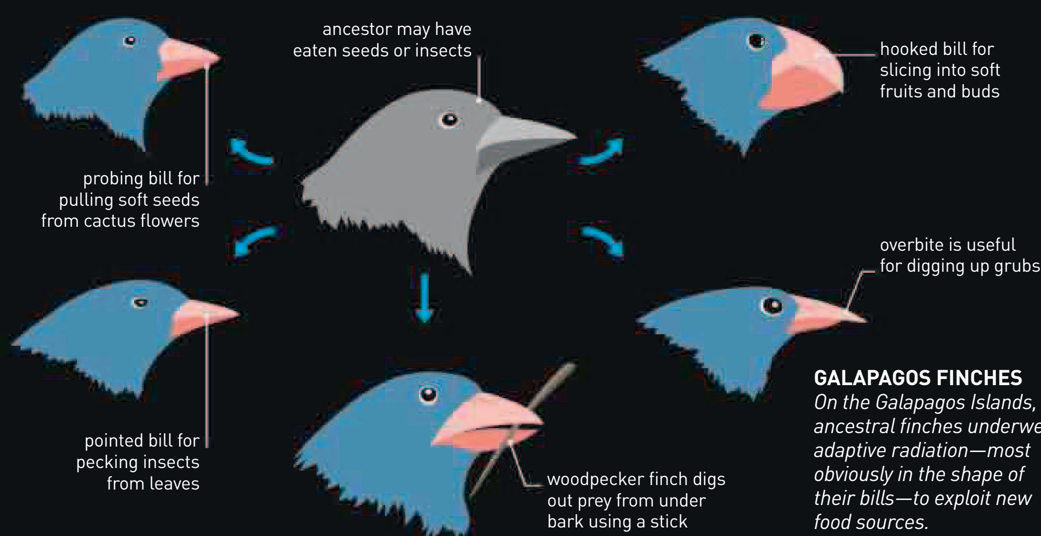


KILLER WHALES AND WHITE SHARKS
These marine predators have both evolved a streamlined shape for greater speed when chasing prey and have a darker upper body and lighter lower body to provide camouflage.

ADAPTIVE RADIATION

When descendents of a common ancestor adapt to different circumstances and diversify, it is known as adaptive radiation. It tends to happen most rapidly in habitats where there are many opportunities for exploiting new ways of life.

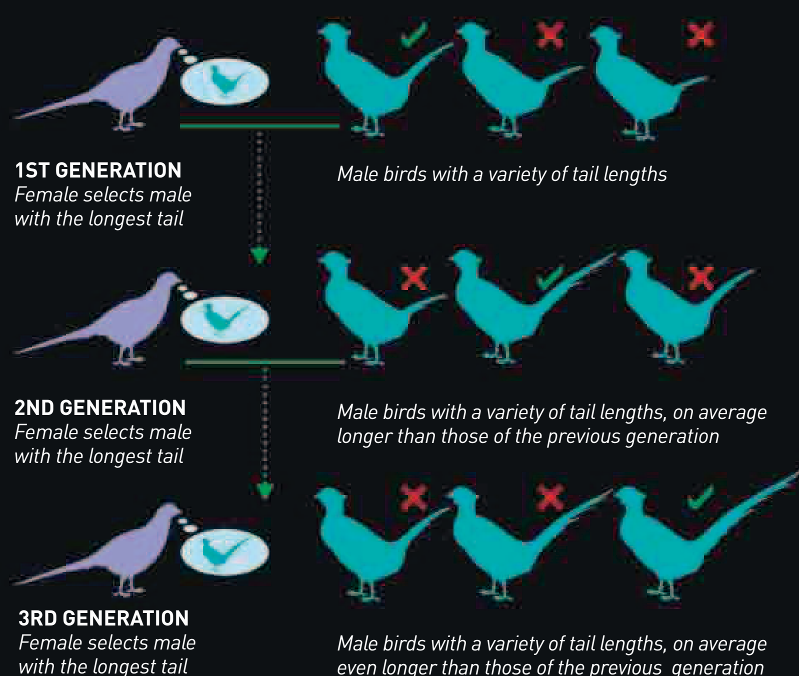
For example, on newly formed islands there may be few competitors and new food sources, so over many generations a pioneering population of one species can diverge to produce many new species, each adapted to a slightly different role.



GALAPAGOS FINCHES
On the Galapagos Islands, ancestral finches underwent adaptive radiation—most obviously in the shape of their bills—to exploit new food sources.

SEXUAL SELECTION

Not all adaptations increase an individual's ability to survive. Sometimes, the selective advantage that drives evolution comes from being better able to attract a mate. For instance, showy plumage can make male birds more vulnerable to predators but it also makes them more successful at courting. As a result, they father more offspring and the genes for showy plumage are passed on.



PHEASANT'S TAIL
Male pheasants with longer tails are more attractive to females, so the genes for a long tail get passed on and male tail length increases with successive generations.